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Inventor(s)

XU; YING

FASTENER DRIVING DEVICE

Abstract

A fastener driving device includes an energy storage unit, an impact unit, a clutch unit, a power unit and a first buffer. The clutch unit includes a guiding mechanism, a driving mechanism, an engaging mechanism and a resetting mechanism. The engaging mechanism drives the driving mechanism to rotate. The engaging mechanism moves away from the driving mechanism for disengagement under a guidance of the guiding mechanism when the engaging mechanism is cooperated with the guiding mechanism, the engaging mechanism moves close to the driving mechanism for reengagement under an action of the resetting mechanism when the engaging mechanism is out of cooperation with the guiding mechanism. The driving mechanism includes a colliding portion, when the engaging mechanism is disengaged from the driving mechanism, an impactor of the impact unit drives the driving mechanism to rotate, the colliding portion strikes the first buffer.

Inventors: XU; YING (SUZHOU, CN)

Applicant: TOGROUP TECHNOLOGY (SUZHOU) CO., LTD. (SUZHOU, CN)

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Background/Summary

FIELD OF THE PRESENT INVENTION

[0001] The present invention relates to a field of mechanical tool products, specifically relating to a type of fastener driving device.

BACKGROUND OF THE PRESENT INVENTION

[0002] For rapid fastening devices (also known as nail guns or fastener driving devices), energy is typically stored by compressing an energy storage unit (e.g., gas, compressed springs, rubber, vacuum, etc.). This stored energy is then rapidly released to perform external work.

[0003] In existing technology, the rapid fastening devices generally use an incomplete gear-rack structure as a driving mechanism, the driving mechanism drives an impact unit to move, the impact unit compresses the energy storage unit to store energy. When the energy storage unit releases energy to drive the impact unit for driving fasteners into an object, the driving mechanism is separated from the impact unit. This leads to a problem that the driving mechanism cannot be correctly cooperated with the impact unit in a condition that the fastener is stuck. This is not conducive to improving the efficiency of the transmission, greatly increasing the time required for the work cycle, and cannot ensure the stable and smooth operation of the impact unit.

SUMMARY OF THE PRESENT INVENTION

[0004] The present invention provides a fastener driving device that operates stably and smoothly.

[0005] Specifically, the present invention provides a fastener driving device including an energy storage unit, an impact unit, a clutch unit and a power unit, the impact unit is connected to the energy storage unit, the impact unit includes an impactor for driving an energy storage unit to store energy and utilizing the energy released by the energy storage unit to drive a fastener into a workpiece; the clutch unit includes a guiding mechanism, a driving mechanism, an engaging mechanism and a resetting mechanism, the driving mechanism is cooperated with the impactor, the engaging mechanism is engaged with the driving mechanism; the power unit includes a rotatable output shaft, the output shaft drives the engaging mechanism to rotate, the engaging mechanism drives the driving mechanism to rotate; the engaging mechanism moves away from the driving mechanism for disengagement under a guidance of the guiding mechanism when the engaging mechanism is cooperated with the guiding mechanism, and the engaging mechanism moves close to the driving mechanism for reengagement under an action of the resetting mechanism when the engaging mechanism is out of cooperation with the guiding mechanism; the fastener driving device further includes a first buffer, the driving mechanism further includes a colliding portion, when the engaging mechanism is disengaged from the driving mechanism, the energy storage unit releases the energy to drive the impactor to move, the impactor drives the driving mechanism to rotate, the colliding portion strikes the first buffer.

[0006] The driving mechanism of the present invention can be properly cooperated with the impact unit, the rotational movement of the driving mechanism being converted into the linear movement of the impact unit can be utilized maximally, ensuring stable and smooth operation of the impact unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a schematic diagram of the fastener driving device of the present invention;

[0008] FIG. 2 is a cross-sectional view along A-A as shown in FIG. 1;

[0009] FIG. 3 is a cross-sectional view along B-B as shown in FIG. 1;

[0010] FIG. 4 is a schematic diagram of a partial structure of the fastener driving device as shown in FIG. 1;

[0011] FIG. 5 is a schematic diagram from another angle of the partial structure of the fastener driving device as described in FIG. 4;

[0012] FIG. 6 is an exploded view of the partial structure as shown in FIG. 4, with the impactor being removed;

[0013] FIG. 7 is a schematic diagram showing a plurality of states in the working cycle of the fastener driving device of the present invention;

[0014] FIG. 8 is another schematic diagram showing a plurality of states in the working cycle of the fastener driving device of the present invention;

[0015] FIG. 9 is a schematic diagram showing a plurality of states of the fastener driving device of the present invention in a condition that the fastener is stuck;

[0016] FIG. 10 is another schematic diagram showing a plurality of states of the fastener driving device of the present invention in a condition that the fastener is stuck.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0017] Here, exemplary embodiments will be explained in detail, which are illustrated in the accompanying drawings. When referring to the drawings in the following description, the same numbers in different drawings represent the same or similar elements, unless otherwise indicated. The embodiments described in the following exemplary embodiments do not represent all embodiments consistent with the present invention.

[0018] The terminology used in the present invention is solely for the purpose of describing specific embodiments and is not intended to limit the present invention. Unless otherwise defined, the technical and scientific terms used in this utility model should be understood in their usual sense by those skilled in the art to which this utility model belongs. The words “first,” “second,” etc., used in the present invention do not indicate any order, quantity, or importance but are merely used to distinguish different components. Similarly, the words “a” or “an” do not indicate a quantity limit, but rather that there exists at least one. “Multiple” or “a plurality of” means two or more. Unless otherwise indicated, terms like “front,” “rear,” “bottom,” and/or “top” are used only for convenience and are not limited to a position or spatial orientation.

[0019] Referencing FIGS. 1 to 10, the fastener driving device of the present invention includes a clutch unit 1, an energy storage unit 2, an impact unit 3, and a power unit 4. The fastener driving device includes a base body 5; the base body 5 includes a mounting space 50, and the clutch unit 1 is mounted in the mounting space 50. Specifically, the base body 5 includes a base 51 and a cover 52 set on the base 51; the base 51 and the cover 52 form the mounting space 50, and the clutch unit 1 is mounted in the mounting space 50.

[0020] The fastener driving device also includes a fastener guiding plate 6 and a fastener storage clip 7. The fastener guiding plate 6 is connected to the base 51, and the fastener storage clip 7 is connected to the fastener guiding plate 6. The fastener storage clip 7 contains a plurality of fasteners (not shown in the FIGS), which are used to be driven into the workpiece by the impact unit 3. The fastener storage clip 7 can feed the plurality of fasteners to the fastener guiding plate 6, and the fastener guiding plate 6 receives the plurality of fasteners from the fastener storage clip 7, the fastener guiding plate 6 performs a guiding function for the fasteners. In this embodiment, the fasteners are nails, the fastener guiding plate 6 is a nail guiding plate, and the fastener storage clip 7 is a nail storage clip.

[0021] The impact unit 3 includes an impactor 30 for driving the energy storage unit 2 to store energy and utilizing the energy released by the energy storage unit 2 to drive the fastener into the workpiece. The impactor 30 is shaped as a long strip, the impactor 30 passes through the mounting space 50 to connect to the energy storage unit 2.

[0022] The clutch unit 1 includes a guiding mechanism 11, a driving mechanism 12, an engaging mechanism 13 and a resetting mechanism 14. The driving mechanism 12 is cooperated with the impactor 30. The engaging mechanism 13 is engaged with the driving mechanism 12. The engaging mechanism 13 moves away from the driving mechanism 12 for disengagement under a guidance of the guiding mechanism 11.

[0023] The power unit 4 includes an electric motor 41, a gear transmitting mechanism 42, and a

rotatable output shaft **43**. The rotational speed and the torque outputted by the electric motor **41** are transmitted to the output shaft **43** via the gear transmitting mechanism **42**. The output shaft **43** is connected to the engaging mechanism **13**, the output shaft **43** drives the engaging mechanism **13** to move. During the rotation of the engaging mechanism **13**, the engaging mechanism **13** moves away from the driving mechanism **12** for disengagement under a guidance of the guiding mechanism **11** when the engaging mechanism **13** is cooperated with the guiding mechanism **11**, and the engaging mechanism **13** moves close to the driving mechanism **12** for reengagement under an action of the resetting mechanism **14** when the engaging mechanism **13** is out of cooperation with the guiding mechanism **11**.

[0024] The gear transmitting mechanism **42** is connected to the base **51** and is used for reducing the rotational speed while increasing the torque. The gear transmitting mechanism **42** includes one stage of planetary gear or a plurality of stages of planetary gear; in this embodiment, the gear transmitting mechanism **42** includes three stages of planetary gear. The output shaft **43** is located between the cover **52** and the base **51**. The base body **5** also includes a first bearing **53** and a second bearing **54**, both ends of the output shaft **43** is respectively supported by the first bearing **53** and the second bearing **54**, specifically, the first end **431** of the output shaft **43** is supported by the first bearing **53**, the first bearing **53** is located between the cover **52** and the first end **431** of the output shaft **43**; the second end **432** of the output shaft **43** is supported by the second bearing **54**, the second bearing **54** is located between the base **51** and the second end **432** of the output shaft **43**. It should be noted that the gear transmitting mechanism **42** is not essential; when the torque outputted by the electric motor **41** is sufficiently large, the shaft of the motor can be directly served as the output shaft **43**.

[0025] The energy storage unit **2** is connected to the base **51** and serves as a medium that can store energy through displacement, such as air springs, mechanical springs, rubber components and vacuum, etc. In this embodiment, the energy storage unit **2** is an air spring, the energy storage unit **2** includes a cylinder **21**, a piston **22** and a sealing ring **23**. The cylinder **21** and the piston **22** form an enclosed space **20** containing gas, a gap between the cylinder **21** and the piston **22** are sealed by the sealing ring **23**. One end of the impactor **30** is connected to the piston **22** through the mounting space **50**, and the other end of the impactor **30** is used to drive the fastener into the workpiece. The impactor **30** can move along with the piston **22**.

[0026] The clutch unit **1** is located between the first bearing **53** and the second bearing **54**. Both the driving mechanism **12** and the engaging mechanism **13** are sleeved on the output shaft **43** and are adjacent to each other, that is, the driving mechanism **12** is adjacent to the engaging mechanism **13** in an axial direction of the output shaft **43**. The engaging mechanism **13** is located between the driving mechanism **12** and the first bearing **53**. The guiding mechanism **11** is fixed in the mounting space **50**. The guiding mechanism **11** is an annular structure, the guiding mechanism **11** is sleeved on the driving mechanism **12**. In this embodiment, the driving mechanism **12** can rotate around the output shaft **43**; the engaging mechanism **13** is sleeved on the output shaft **43** and is movable along an axial direction of the output shaft, the engaging mechanism **13** can move relative to the driving mechanism **12** along the axial direction of the output shaft **43**. The output shaft **43** drives the engaging mechanism **13** to rotate, that is to say, the engaging mechanism **13** rotates with the output shaft **43**.

[0027] The resetting mechanism **14** is located on a side, away from the driving mechanism **12**, of the engaging mechanism **13**; the resetting mechanism **14** presses against the engaging mechanism **13**. Specifically, the resetting mechanism **14** is mounted between the first bearing **53** and the engaging mechanism **13**. One end of the resetting mechanism **14** presses against the engaging mechanism **13**, and the other end of the resetting mechanism **14** presses against the first bearing **53**. In this embodiment, the resetting mechanism **14** is an elastic resetting member, specifically, the elastic resetting member is a spring, the spring is sleeved on the output shaft **43**. One end of the spring presses against the engaging mechanism **13**, and the other end of the spring presses against

the first bearing **53**.

[0028] The guiding mechanism **11** includes at least one guiding portion, the driving mechanism **12** includes at least one engaging tooth, and the engaging mechanism **13** includes at least one engaging portion and at least one cooperating portion. The output shaft **43** drives the engaging mechanism **13** to rotate. During the rotation of the engaging mechanism **13**, the at least one engaging portion is engaged with the at least one engaging tooth, so that the engaging mechanism **13** drives the driving mechanism **12** to rotate; the at least one cooperating portion is cooperated with the at least one guiding portion, so that the at least one engaging portion moves away from the at least one engaging tooth for disengagement under a guidance of the at least one guiding portion.

[0029] Specifically, the guiding mechanism **11** includes a main body **110**, a first guiding portion **111** and a second guiding portion **112**; the first guiding portion **111** and the second guiding portion **112** are arranged on the main body **110**. Both the first guiding portion **111** and the second guiding portion **112** are located on a side, close to the engaging mechanism **13**, of the main body **110**. The first guiding portion **111** has a first mating surface **S1**, and the second guiding portion **112** has a second mating surface **S2**. Both the first mating surface **S1** and the second mating face **S2** face towards the engaging mechanism **13**.

[0030] The engaging mechanism **13** includes a base portion **130**, a first engaging portion **131** extending from one side of the base portion **130**, a first cooperating portion **133** extending from the first engaging portion **131**, a second engaging portion **132** extending from the other side of the base portion **130**, and a second cooperating portion **134** extending from the second engaging portion **132**. The first cooperating portion **133** and the second cooperating portion **134** extend in a direction that is perpendicular or nearly perpendicular to the axial direction of the output shaft **43**, the first engaging portion **131** and the second engaging portion **132** extend in a direction that is perpendicular or nearly perpendicular to the axial direction of the output shaft **43**.

[0031] In the radial direction of the output shaft **43**, the first engaging portion **131** is closer to the base portion **130** than the first cooperating portion **133**, and the second engaging portion **132** is closer to the base portion **130** than the second cooperating portion **134**. The base portion **130** is sleeved on the output shaft **43**. The radial direction of the output shaft **43** is perpendicular to the axial direction of the output shaft **43**. The first cooperating portion **133** has a third mating surface **S3**, and the second cooperating portion **134** has a fourth mating surface **S4**. Both the third mating surface **S3** and the fourth mating surface **S4** face towards the guiding mechanism **11**. The third mating surface **S3** of the first cooperating portion **133** is cooperated with the first mating surface **S1** of the first guiding portion **111**, and the fourth mating surface **S4** of the second cooperating portion **134** is cooperated with the second mating surface **S2** of the second guiding portion **112**.

[0032] In this embodiment, the first guiding portion **111** and the second guiding portion **112** are arranged on different circumferences centered on a point on the central axis of the output shaft **43**; the first cooperating portion **133** and the second cooperating portion **134** are also arranged on different circumferences centered on a point on the central axis of the output shaft **43**. In the radial direction of the output shaft **43**, the first guiding portion **111** and the second guiding portion **112** are corresponding to each other, the first cooperating portion **133** and the second cooperating portion **134** are corresponding to each other.

[0033] During the rotation of the engaging mechanism **13** with the output shaft **43**, the relationship between the mating surface of the cooperating portion and the mating surface of the guiding portion is changed from “begin to cooperate” to “out of cooperation”. The detailed process is as follows: The mating surface of the cooperating portion and the mating surface of the guiding portion begin to cooperate, and under the guidance of the mating surface of the guiding portion, the engaging mechanism **13** moves away from the driving mechanism **12** in the axial direction of the output shaft **43**. The engaging mechanism **13** is disengaged from the driving mechanism **12**, and the engaging mechanism **13** continues to rotate with the output shaft **43**, the mating surface of the cooperating portion is out of cooperation with the mating surface of the guiding portion. In this

embodiment, the mating surfaces are spiral surfaces, and the helix angle of the spiral surface of the guiding portion is not greater than 60°.

[0034] Preferably, each of the first mating surface **S1** and the second mating surface **S2** has a first edge close to the output shaft **43** and a second edge far away from the output shaft **43**. The distance **D1** between the second edge of the first mating surface **S1** and the output shaft **43** is less than the distance **D2** between the first edge of the second mating surface **S2** and the output shaft **43**. When the first cooperating portion **133** is rotated to a position which is corresponding to the position of the second guiding portion **112** along the radial direction of the output shaft **43**, the third mating surface **S3** of the first cooperating portion **133** is not cooperated with the second mating surface **S2** of the second guiding portion **112**, and the fourth mating surface **S4** of the second cooperating portion **134** is not cooperated with the first mating surface

[0035] **S1** of the first guiding portion **111**. Thus, as the engaging mechanism **13** rotates one full circle with the output shaft **43**, the impact unit **3** only achieves a complete reciprocating movement, i.e., the impact unit **3** achieves one driving action, avoiding too frequent driving actions from affecting stability of the fastener driving device.

[0036] The driving mechanism **12** includes a bracket **120** and a plurality of pins **129** mounted on the bracket **120**, the bracket **120** is rotatably sleeved on the output shaft **43**. The bracket **120** is generally shaped as a disk. The plurality of pins **129** are arranged in a circumferential direction. The bracket **120** includes a first sub-bracket **123** and a second sub-bracket **124** which are corresponding to each other along the axial direction of the output shaft **43**. The first sub-bracket **123** is closer to the engaging mechanism **13** than the second sub-bracket **124**. The first sub-bracket **123** includes a plurality of first pin holes **125**. The second sub-bracket **124** includes a plurality of second pin holes **126**. Both ends of the pin **129** are respectively inserted into the first pin hole **125** and the second pin hole **126** which are corresponding to each other along the axial direction of the output shaft **43**. The impactor **30** includes a plurality of driving teeth **31**, the plurality of driving teeth **31** are engaged with the plurality of pins **129**. The driving mechanism **12** further includes a pad **127**, the pad **127** is located on a side, close to the engaging mechanism **13**, of the bracket **120**. The pad **127** presses against the bracket **120** and covers at least a part of an area of the first pin hole **125**, to prevent the pin **129** from falling out of the first pin hole **125**. In this embodiment, the pin and the pin hole are designed for clearance fit, so a pad **127** is needed to prevent the pin **129** from falling out of the first pin hole **125**. It should be noted that the pad **127** is not mandatory; the pad **127** can be omitted if the pin and the pin hole are designed for interference fit. In this embodiment, the pin and the pin hole are designed for clearance fit, to reduce the friction between the pins **129** and the driving teeth **31**.

[0037] The driving mechanism **12** also includes a first engaging tooth **121** and a second engaging tooth **122** located on the bracket **120**. The first engaging tooth **121** and the second engaging tooth **122** face towards the engaging mechanism **13**. Specifically, the first engaging tooth **121** is engaged with the first engaging portion **131** of the engaging mechanism **13**, and the second engaging tooth **122** is engaged with the second engaging portion **132** of the engaging mechanism **13**. In detail, “engagement between the engaging portion and the engaging tooth” means “the side face of the engaging portion presses against the side face of the engaging tooth”. Specifically, the engaging tooth includes a first engaging surface, and the engaging portion includes a second engaging surface, the first engaging surface is cooperated with the second engaging surface. Both the first engaging surface and the second engaging surface are perpendicular to a plane of rotation of the bracket **120**, the plane of rotation of the bracket **120** is perpendicular to a plane where the center axis of the output shaft **43** is located. The output shaft **43** drives the engaging mechanism **13** to rotate. When the engaging portions are engaged with the engaging teeth, the engaging mechanism **13** drives the driving mechanism **12** to rotate, when the pins **129** are engaged with the driving teeth **31**, the driving mechanism **12** drives the impactor **30** to move along a first direction, the first direction is a direction which is towards the energy storage unit **2**, the impactor compresses the

piston **22**, then energy can be stored by the energy storage unit **2**. In this embodiment, via the engagement between the pins **129** of the driving mechanism **12** and the driving teeth **31** of the impactor **30**, a rotational movement of the driving mechanism **12** is transformed into a linear movement of the impactor **30**.

[0038] In the present invention, the output shaft **43** rotates counterclockwise to drive the engaging mechanism **13** to rotate counterclockwise. The relationship between the mating surfaces (i.e., the third mating surface **S3** and the fourth mating surface **S4**) of the cooperating portions (i.e., the first cooperating portion **133** and the second cooperating portion **134**) of the engaging mechanism **13** and the mating surfaces (i.e., the first mating surface **S1** and the second mating surface **S2**) of the guiding portions (i.e., the first guiding portion **111** and the second guiding portion **112**) of the guiding mechanism **11** is changed from “begin to cooperate” to “out of cooperation”, during a process of “begin to cooperate” to “out of cooperation”, the engaging mechanism **13** moves away from the driving mechanism **12** along the axial direction of the output shaft **43** until the engaging portions of the engaging mechanism **13** are about to be disengaged from the engaging teeth of the driving mechanism **12**, the output shaft **43** continues to rotate counterclockwise, the engaging portions of the engaging mechanism **13** are disengaged from the engaging teeth of the driving mechanism **12**.

[0039] When the engaging portions of the engaging mechanism **13** are disengaged from the engaging teeth of the driving mechanism **12**, under an action of the energy storage unit **2** (i.e., the pressure from the energy storage unit **2** acts on the impactor **30**), the impactor **30** drives the driving mechanism **12** to move rapidly (i.e., the impactor **30** moves away from the energy storage unit **2**, while the driving mechanism **12** rotates clockwise). The impactor **30** drives the fasteners received in the fastener guiding plate **6** into the workpiece (i.e., typically wood, metal plates or concrete). The driving mechanism **12** returns to its initial position. The output shaft **43** continues to rotate an angle in a counterclockwise direction, the cooperating portions (i.e., the first cooperating portion **133** and the second cooperating portion **134**) of the engaging mechanism **13** are disengaged from the guiding portions (i.e., the first guiding portion **111** and the second guiding portion **112**) of the guiding mechanism **11**, under an action of the resetting mechanism, the engaging mechanism **13** moves close to the driving mechanism **12** along the axial direction of the output shaft **43** to return to its initial position. In this embodiment, the output shaft **43** includes a shoulder **430** as a stopping portion, under the action of the resetting mechanism, the engaging mechanism **13** moves close to the driving mechanism **12** until the engaging mechanism **13** presses against the shoulder **430** of the output shaft **43**.

[0040] The fastener driving device also includes a first buffer **8** mounted between the base **51** and the driving mechanism **12**. This first buffer **8** is located on a side, away from the engaging mechanism **13**, of the bracket **120**. The driving mechanism **12** further includes a colliding portion **128**; the colliding portion **128** is located on a side, away from the engaging mechanism **13**, of the bracket **120**. During the rotation of the driving mechanism **12** (i.e., the driving mechanism **12** is driven by the impactor **30**), the colliding portion **128** will strike the first buffer **8**.

[0041] Specifically, when the impactor **30** moves away from the energy storage unit **2** along a second direction and drives the driving mechanism **12** to rotate in a clockwise direction, the driving mechanism **12** carries a certain amount of rotational energy. When the impactor **30** completes a driving action for the fasteners, the residual rotational energy of the driving mechanism **12** will be absorbed or dissipated by the first buffer **8**, in this design, the driving mechanism **12** and/or the impactor **30** can be prevented from being damaged. The first buffer **8** is located on a side, away from the engaging mechanism **13**, of the bracket **120**; at least part of the first buffer **8** is distributed along a circumference of the bracket **120**. In this design, it is convenient for the first buffer **8** to absorb or dissipate the residual rotational energy of the driving mechanism **12**.

[0042] The first buffer **8** is generally made of rubber or other elastic material such as thermoplastic elastomer, and the first buffer **8** can also be a spring.

[0043] The fastener driving device also includes a second buffer **9** mounted between the base **51** and the piston **22**. During a process of the impactor **30** driving the energy storage unit **2** to store energy, the impactor **30** pushes the piston **22** to move away from the second buffer **9**. During a process of the impactor **30** being subjected to the energy released by the energy storage unit **2** to drive the fasteners into the workpiece, the piston **22** pushes the impactor **30** to move in a direction which is towards the second buffer **9** until the piston **22** strikes the second buffer **9**. When the impactor **30** completes the driving action for the fastener, the residual energy of the piston **22** is absorbed or dissipated by the second buffer **9**. In this design, the driving mechanism **12** and/or the impactor **30** can be prevented from being damaged.

[0044] When the impactor **30** completes the driving action for the fastener, the output shaft **43** continues to rotate an angle in a counterclockwise direction, the cooperating portions (i.e., the first cooperating portion **133** and the second cooperating portion **134**) of the engaging mechanism **13** are disengaged from the guiding portions (i.e., the first guiding portion **111** and the second guiding portion **112**) of the guiding mechanism **11**, under the action of the resetting mechanism **14**, the engaging mechanism **13** moves close to the driving mechanism **12** along the axial direction of the output shaft **43**, to return to its initial position. At this time, because the engaging portions of the engaging mechanism **13** are not engaged with the engaging teeth of the driving mechanism **12**, the colliding portion **128** of the driving mechanism **12** and the piston **22** of the energy storage unit **2** are respectively parked at the first buffer **8** and the second buffer **9**.

[0045] As shown in FIGS. **7** and **8**, states a, b, c, d, e are corresponding to a working cycle. The output shaft **43** rotates counterclockwise, the output shaft **43** drives the engaging mechanism **13** to rotate, the engaging mechanism **13** drives the driving mechanism **12** to rotate counterclockwise to the position shown in state a. At this time, the third pin of the driving mechanism **12** is about to be disengaged from the third driving tooth of the impactor **30**, and the fourth pin is about to be engaged with the fourth driving tooth. As the output shaft **43** continues to rotate counterclockwise, the fourth pin is engaged with the fourth driving tooth. At this time, the end, close to the piston **22**, of the impactor **30** is located at the position shown by the dotted line in FIG. **3**. Meanwhile, the mating surfaces (i.e., the third mating surface **S3** and the fourth mating surface **S4**) of the cooperating portions (i.e., the first cooperating portion **133** and the second cooperating portions **134**) of the engaging mechanism **13** begin to cooperate with the mating surfaces (i.e., the first mating surface **S1** and second mating surface **S2**) of the guiding portions (i.e., the first guiding portion **111** and the second guiding portion **112**) of the guiding mechanism **11**. The engaging mechanism **13** moves away from the driving mechanism **12** along the axial direction of the output shaft **43** until the engaging portions of the engaging mechanism **13** are about to be disengaged from the engaging teeth of the driving mechanism **12**, as shown in state b. The output shaft **43** continues to rotate counterclockwise, the engaging portions of the engaging mechanism **13** are disengaged from the engaging teeth of the driving mechanism **12**. The impactor **30** moves away from the energy storage unit **2** in a second direction under the action of the piston **22**, the impactor **30** drives the driving mechanism **12** to rotate, the impactor **30** and the driving mechanism **12** move together at a high speed. Specifically, the driving mechanism **12** rotates in a clockwise direction driven by the impactor **30**, and the impactor **30** drives the fasteners received in the fastener guiding plate **6** into the workpiece. The piston **22** strikes the second buffer **9**, while the colliding portion **128** of the driving mechanism **12** strikes the first buffer **8**. The output shaft **43** continues to rotate an angle in a counterclockwise direction, and the first cooperating portion **133** and the second cooperating portion **134** of the engaging mechanism **13** are disengaged from the first guiding portion **111** and second guiding portion **112** of the guiding mechanism **11**. Under the action of the resetting mechanism **14**, the engaging mechanism **13** moves close to the driving mechanism **12** along the axial direction of the output shaft **43** to return to its initial position, as shown in state c. At this time, because the engaging portions of the engaging mechanism **13** are not engaged with the engaging teeth of the driving mechanism **12**, the colliding portion **128** of the driving mechanism **12** and the

piston **22** of the energy storage unit **2** are respectively parked at the first buffer **8** and the second buffer **9**.

[0046] The output shaft **43** continues to rotate counterclockwise, as shown in state d, the engaging portions of the engaging mechanism **13** are re-engaged with the engaging teeth of the driving mechanism **12**, so that the engaging mechanism **13** drives the driving mechanism **12** to rotate, the engaging mechanism **13** and the driving mechanism **12** rotate together. The output shaft **43** continues to rotate counterclockwise, as shown in state e, the engaging mechanism **13** drives the driving mechanism **12** to rotate, and the pins **129** of the driving mechanism **12** are engaged with the driving teeth **31** of the impactor **30** in turn, so that the driving mechanism **12** drives the impactor **30** to move close to the energy storage unit **2**, thereby compressing the piston **22** to store energy. In this process, the rotational movement of the driving mechanism **12** is converted into the linear movement of the impactor **30**. The output shaft **43** continues to rotate counterclockwise until the engaging mechanism **13** drives the driving mechanism **12** rotate counterclockwise to the position shown in state a.

[0047] As shown in FIGS. **9** and **10**, for some reason, the fastener is stuck in the fastener guiding plate **6**, the impactor **30** may stop anywhere along the second direction, as shown in state f. In the process of the driving mechanism **12** driving the impactor **30** to move, it is necessary that at least two pins **129** of the driving mechanism **12** are engaged with at least two driving teeth **31** of the impactor **30**, if the pins **129** of the driving mechanism **12** are disengaged from the driving teeth **31** of the impactor **30** when the impactor **30** is moving along the second direction, in a situation that the fastener is stuck, the pins **129** of the driving mechanism **12** may not be correctly re-engaged with the driving teeth **31** of the impactor **30** due to the randomness of the stopping position of the impactor **30**. In the present invention, the engagement between the driving teeth **31** of the impactor **30** and the pins **129** of the driving mechanism **12** is always maintained, therefor, even if the fastener is stuck, it is ensured that the pins **129** of the driving mechanism **12** and the driving teeth **31** of the impactor **30** will be correctly engaged.

[0048] When the fastener is stuck, the output shaft **43** continues to rotate counterclockwise, the cooperating portions (i.e., the first cooperating portion **133** and the second cooperating portions **134**) of the engaging mechanism **13** are disengaged from the guiding portions (i.e., the first guiding portion **111** and the second guiding portion **112**) of the guiding mechanism **11**. Under the action of the resetting mechanism **14**, the engaging mechanism **13** moves close to the driving mechanism **12** along the axial direction of the output shaft **43** to return to its initial position. The engaging mechanism **13** continues to rotate with the output shaft **43**, and the engaging portions of the engaging mechanism **13** are engaged with the engaging teeth of the driving mechanism **12**, as shown in state g. The output shaft **43** continues to rotate counterclockwise, the output shaft **43** drives the engaging mechanism **13** to rotate, the engaging mechanism **13** drives the driving mechanism **12** to rotate. The driving mechanism **12** can drive the impactor **30** to compress the piston **22**, the impactor **30** and the driving mechanism **12** can return to the initial state of the working cycle, as shown in state a.

[0049] In the present invention, when the impactor **30** moves along the second direction, the driving teeth **31** of the impactor **30** will not be disengaged from the pins **129** of the driving mechanism **12**, even if the fastener is stuck, it is ensured that the driving teeth **31** of the impactor **30** are correctly engaged with the pins **129** of the driving mechanism **12**. Moreover, during the energy storage process of the energy storage unit **2**, the utilization of the rotational movement of the driving mechanism **12** is maximized, that is, the rotational movement of the driving mechanism **12** is maximally converted into the linear movement of the impactor **30**. In addition, a quick release of the energy stored in the energy storage unit **2** can be achieved, the stability and smoothness of the operation of the impactor **30** can be ensured, the quality of the work can be improved, and the time required for the working cycle can be reduced. Even if the fastener is stuck, it can be quickly return to the initial state of the working cycle, it is more convenient to use the fastener driving

device.

[0050] The above description is only the preferred embodiment of the present invention and is not intended to limit the present invention in any form. Although the present invention has been disclosed as above in the preferred embodiment, it is not intended to limit the application. Any simple modifications, equivalent changes, and modifications made to the above embodiments by anyone skilled in the art, without departing from the technical scope of the present invention, still fall within the scope of the present invention.

Claims

1. A fastener driving device, comprising: an energy storage unit; an impact unit, connected to the energy storage unit, the impact unit comprising an impactor for driving an energy storage unit to store energy and utilizing the energy released by the energy storage unit to drive a fastener into a workpiece; a clutch unit, comprising a guiding mechanism, a driving mechanism, an engaging mechanism and a resetting mechanism, the driving mechanism being cooperated with the impactor, the engaging mechanism being cooperated with the driving mechanism; a power unit, comprising a rotatable output shaft, the output shaft driving the engaging mechanism to rotate, the engaging mechanism driving the driving mechanism to rotate; wherein the engaging mechanism moves away from the driving mechanism for disengagement under a guidance of the guiding mechanism when the engaging mechanism is cooperated with the guiding mechanism, and the engaging mechanism moves close to the driving mechanism for reengagement under an action of the resetting mechanism when the engaging mechanism is out of cooperation with the guiding mechanism; the fastener driving device further comprises a first buffer, the driving mechanism further comprises a colliding portion, when the engaging mechanism is disengaged from the driving mechanism, the energy storage unit releases the energy to drive the impactor to move, the impactor drives the driving mechanism to rotate, the colliding portion strikes the first buffer.
2. The fastener driving device according to claim 1, wherein the fastener driving device comprises a base body, the base body comprises a mounting space, the clutch unit is mounted in the mounting space, the impactor passes through the mounting space to connect to the energy storage unit.
3. The fastener driving device according to claim 2, wherein the driving mechanism is sleeved on the output shaft, the guiding mechanism is fixedly mounted in the mounting space and is sleeved on the driving mechanism, the engaging mechanism is sleeved on the output shaft and is movable along an axial direction of the output shaft, the driving mechanism is adjacent to the engaging mechanism in the axial direction of the output shaft.
4. The fastener driving device according to claim 3, wherein the driving mechanism comprises a bracket and a plurality of pins mounted on the bracket, the bracket is rotatably sleeved on the output shaft, the bracket comprises a first sub-bracket and a second sub-bracket corresponding to each other, the first sub-bracket comprises a first pin hole, the second sub-bracket comprises a second pin hole, both ends of the pin are mounted in the first pin hole and the second pin hole respectively, the impactor comprises a plurality of driving teeth, the driving teeth are used for being engaged with the pins.
5. The fastener driving device according to claim 4, wherein the driving mechanism further comprises a pad, the pad presses against the bracket and covers at least a part of an area of the first pin hole, to prevent the pin from being separated from the first pin hole.
6. The fastener driving device according to claim 4, wherein the plurality of pins are distributed in a circumferential direction.
7. The fastener driving device according to claim 5, wherein the first sub-bracket and the second sub-bracket are corresponding to each other along the axial direction of the output shaft, the first sub-bracket is closer to the engaging mechanism relative to the second sub-bracket, the pad is located on a side, close to the engaging mechanism, of the bracket.

8. The fastener driving device according to claim 4, wherein the driving mechanism comprises at least one engaging tooth disposed on the bracket, the at least one engaging tooth faces the engaging mechanism; the engaging mechanism comprises at least one engaging portion and at least one cooperating portion, the guiding mechanism comprises at least one guiding portion, the output shaft drives the engaging mechanism to rotate, during a rotation of the engaging mechanism, the at least one engaging portion is engaged with the at least one engaging tooth, so that the engaging mechanism drives the driving mechanism to rotate; the at least one cooperating portion is cooperated with the at least one guiding portion, so that the at least one engaging portion moves away from the at least one engaging tooth for disengagement under a guidance of the at least one guiding portion.

9. The fastener driving device according to claim 8, wherein each of the guiding portion and the cooperating portion comprises a mating surface, during the rotation of the engaging mechanism, the mating surface of the guiding portion is cooperated with the mating surface of the cooperating portion, so that the engaging portion moves away from the engaging tooth for disengagement under a guidance of the guiding portion.

10. The fastener driving device according to claim 9, wherein the mating surface is a spiral surface.

11. The fastener driving device according to claim 8, wherein the at least one guiding portion comprises at least two guiding portions, the at least two guiding portions are arranged on different circumferences centered on a point on a center axis of the output shaft; the at least one cooperating portion comprises at least two cooperating portions, the at least two cooperating portions are arranged on different circumferences centered on a point on the center axis of the output shaft.

12. The fastener driving device according to claim 11, wherein the at least two guiding portions comprise a first guiding portion and a second guiding portion, the first guiding portion and the second guiding portion are corresponding to each other along a radial direction of the output shaft; the at least two cooperating portions comprise a first cooperating portion and a second cooperating portion, the first cooperating portion and the second cooperating portion are corresponding to each other along the radial direction of the output shaft.

13. The fastener driving device according to claim 2, wherein the resetting mechanism is located on a side, away from the driving mechanism, of the engaging mechanism; and the resetting mechanism presses against the engaging mechanism, one end of the resetting mechanism presses against the engaging mechanism, and the other end of the resetting mechanism presses against the base body; the resetting mechanism is an elastic resetting member.

14. The fastener driving device according to claim 3, wherein the resetting mechanism is a spring, the spring is sleeved on the output shaft, the spring is located on a side, away from the driving mechanism, of the engaging mechanism; one end of the spring presses against the engaging mechanism, and the other end of the spring presses against the base body.

15. The fastener driving device according to claim 4, wherein the first buffer is mounted in the mounting space, the first buffer and the colliding portion are located on a side, away from the engaging mechanism, of the bracket.

16. The fastener driving device according to claim 15, wherein at least a part of the first buffer is distributed along a circumference of the bracket.

17. The fastener driving device according to claim 2, wherein the energy storage unit comprises a cylinder, a piston and a sealing ring; the cylinder and the piston form an enclosed space containing gas, and a gap between the cylinder and the piston are sealed by the sealing ring; when the engaging mechanism is disengaged from the driving mechanism, the piston pushes the impactor to move.

18. The fastener driving device according to claim 17, wherein the fastener driving device further comprises a second buffer mounted in the base body, the piston pushes the impactor to move and strikes the second buffer.

19. The fastener driving device according to claim 8, wherein the engaging tooth comprises a first

engaging surface, and the engaging portion comprises a second engaging surface, the first engaging surface is engaged with the second engaging surface, the first engaging surface and the second engaging surface are perpendicular to a plane of rotation of the bracket, the plane of rotation of the bracket is perpendicular to a plane where a center axis of the output shaft is located.

20. The fastener driving device according to claim 10, wherein a helix angle of the spiral surface of the guiding portion is not greater than 60° .
